

Epic 4: Model Building

Story 4: XGBoost Model

The XGBoost (Extreme Gradient Boosting) algorithm is implemented to achieve high-accuracy flood prediction by using an advanced ensemble learning technique. XGBoost is known for its speed, efficiency, and excellent predictive performance, making it one of the most widely used algorithms for classification problems.

XGBoost Algorithm

The XGBoost Classifier is initialized and trained using the training dataset. It builds multiple decision trees sequentially, where each new tree learns from the errors made by the previous trees. This boosting process improves prediction accuracy and reduces overfitting.

Model Training

The model is trained using the `fit()` method with the training dataset (`X_train` and `y_train`). During training, XGBoost learns the complex relationships between weather-related features and flood occurrence, producing a highly optimized predictive model.

Prediction

After training, the `predict()` method is used to generate flood predictions for the testing dataset (`X_test`). The predicted values are then compared with the actual target values (`y_test`) to evaluate the model's effectiveness.

Model Evaluation

The performance of the XGBoost model is assessed using the following evaluation metrics:

- **Accuracy Score** – Measures the percentage of correctly classified instances.
- **Confusion Matrix** – Displays the number of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).
- **Classification Report** – Provides detailed performance metrics including Precision, Recall, F1-Score, and Support for each class.

Outcome

- Successfully trained the XGBoost classifier.
- Generated accurate flood predictions using the testing dataset.
- Evaluated the model using Accuracy, Confusion Matrix, Precision, Recall, and F1-Score.

- Achieved the highest prediction accuracy (96.55%) among all the implemented machine learning models.
- Selected the XGBoost model as the final model for deployment in the Flask-based AI Flood Prediction System.

Figure 15: XGBoost model training and performance evaluation using the flood prediction dataset.

```
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.metrics import confusion_matrix, classification_report, accuracy_score

def xgboost(X_train, X_test, y_train, y_test):
    print("\n===== XGBOOST MODEL BUILDING =====")

    # Initialize Gradient Boosting Classifier
    model = GradientBoostingClassifier()

    # Train the model using training data
    model.fit(X_train, y_train)
    print("[INFO] XGBoost model training completed.")

    # Predict on test data
    y_pred = model.predict(X_test)
    print("[INFO] Prediction completed on test data.")

    # Evaluate the model
    accuracy = accuracy_score(y_test, y_pred)
    cm = confusion_matrix(y_test, y_pred)
    cr = classification_report(y_test, y_pred)

    # Display results
    print("\n[RESULT] Accuracy:", accuracy)
    print("\nConfusion Matrix:")
    print(cm)
    print("\nClassification Report:")
    print(cr)

    # Return model and predictions
    return model, y_pred

# Example Usage
xgb_model, xgb_pred = xgboost(X_train, X_test, y_train, y_test)
```